

Project 01 - Superstore Sales Analysis

Executive Summary

Analysis of the Superstore dataset using SQL, focused on sales, profitability, margins, regional/segment performance, and data quality investigation.

Key Results

Metric	Result
Total sales	\$2,297,200.86
Total profit	\$286,397.02
Overall margin	12.47%
Best category margin	Technology — 17.40%
Weakest category margin	Furniture — 2.49%

Business Findings

- Furniture was the main profitability concern, with a margin of approximately 2.49%.
- Tables and Bookcases were identified as loss-making subcategories.
- The Central region had the weakest overall margin, approximately 7.92%.
- Home Office showed the strongest customer-segment margin, approximately 14.03%.

Data Quality Case Studies

Synthetic quality issues were introduced on a test copy of the dataset to demonstrate a professional detection, investigation, correction, and validation workflow. The source dataset itself was not represented as having these errors.

A duplicate-record scenario was also simulated. The test table temporarily contained duplicate copies that inflated sales by \$656.76 (0.22%) and profit by \$100.96 (0.04%). The duplicate copies were then removed while retaining one valid record.

Advanced Incident Case

A synthetic database-outage scenario was designed around a recovery burst of 10,000 queued requests. The analyst role is to quantify request volume, errors, latency and retry behavior, then recommend controls such as rate limiting, bounded queues, connection pooling, backpressure, bounded retries, circuit breakers and query optimization.